

# Exploiting ASP.NET TemplateParser — Part II: SharePoint (CVE-2023-33160)

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## Content

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# Exploiting ASP.NET TemplateParser — Part II: SharePoint (CVE-2023-33160)

In [Part I](#), we dug into the internals of the ASP.NET `TemplateParser` and elaborated its capabilities in respect to exploitation.

In this part, we will look into whether and how this can also be exploited to gain Remote Code Execution. While this research was originally focussed on the `TemplateParser`, the newly discovered technique was also applicable to SharePoint on-premises and SharePoint Online. So we'll elaborate on how SharePoint protects against the use of malicious code and will present a novel trick that allowed to bypass these security measures ([CVE-2023-33160](#)).

## Introduction

SharePoint users are allowed to upload and create custom `.aspx` pages:

A fundamental assumption of the Windows SharePoint Services technology is that “untrusted users” can upload and create ASPX pages within the system on which Windows SharePoint Services is running. These users should be prevented from adding server-side code within ASPX pages, but there should be a list of approved controls that those untrusted users can use. One way to provide these controls is to create a Safe Controls list.

— [https://learn.microsoft.com/en-us/previous-versions/office/developer/sharepoint-2007/ms581321\(v=office.12\)](https://learn.microsoft.com/en-us/previous-versions/office/developer/sharepoint-2007/ms581321(v=office.12))

For that, there is the `SPPageParserFilter` class in SharePoint that extends the `PageParserFilter` class of ASP.NET to validate the server control type specified in the source code against this Safe Controls list. This validation happens already during the tokenization step in the `TemplateParser.ProcessBeginTag(Match, string)` method:

```
// If we have a control type filter, make sure the child control is allowed
if (_pageParserFilter != null) {
    Debug.Assert(childType != null);

    if (!_pageParserFilter.AllowControlInternal(childType, subBuilder)) {
        ProcessError(SR.GetString(SR.Control_type_not_allowed, childType.FullName));
        return true;
    }
}
```

In this code excerpt the `childType` variable represents the `Type` of the specified server control. The inner workings of SharePoint's `SPPageParserFilter` were previously elaborated by Soroush Dalili in *A Security Review of SharePoint Site Pages* as well as in *Room for Escape: Scribbling Outside the Lines of Template Security* by Oleksandr Mirosh and Alvaro Muñoz. We recommend reading their work first to be on the same page.

## TemplateParser Challenges and Ideas

As detailed in [Part I](#), the parsing step of the `TemplateParser` has the following policy:

-

`Type` can only be controlled using custom server controls and the `@ Register` directive:

```
<%@ Register
    TagPrefix="MyPrefix"
    Assembly="MyAssembly"
    Namespace="MyNamespace"
%>
<MyPrefix:MyTypeName
    runat="server"
/>
```

-

Server control tag names can only be `[\w: . ]+`

-

Server control type resolution is done using

```
Assembly.Load("MyAssembly").GetType("MyNamespace" + "." + "MyTypeName")
```

-

Property types are derived from the specified server control type by reflection

Let's look into the challenges one by one.

## Controlling The Property Type

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Property types are derived from the specified server control type by reflection

Idea: Use a generic type where the type argument defines the property's type.

Consider the [generic class](#) `ExpandedWrapper<TExpandedElement>` where the generic type parameter `TExpandedElement` determines the type of the `ExpandedElement` property:

```
typeof(System.Data.Services.Internal.ExpandedWrapper<DateTime>)
    .GetProperty("ExpandedElement").PropertyType == typeof(DateTime)
```

This also means, assigning a value to the `ExpandedElement` property of a `ExpandedWrapper<DateTime>` instance would result in the `DateTimeConverter.ConvertFromInvariantString(string)` method being called.

But how can we create an instance of `ExpandedWrapper<DateTime>` ?

## Controlling The Custom Server Control Type

---

Server control type resolution is done using

```
Assembly.Load("MyAssembly").GetType("MyNamespace" + "." + "MyTypeName")
```

Idea: Maybe we could trick .NET into ignoring the appendix, similar to what we have observed in [Bypassing .NET Serialization Binders](#).

What if we could specify the targeted type in the `Namespace` entirely and somehow hide the appended `"." + "MyTypeName"` portion?

Let's make a quick test and insert an arbitrary character right after the namespace and see whether the type can still be resolved:

```
C# Interactive
1 > #r "C:\Windows\Microsoft.NET\assembly\GAC_MSIL\System.Data.Services\v4.0.0.0_b77a5c561934e089
\System.Data.Services.dll"
1 > var targetType = typeof(System.Data.Services.Internal.ExpandedWrapper<System.DateTime>);
2 . var myAssembly = targetType.Assembly.FullName;
3 . var myNamespace = targetType.FullName;
1 > Enumerable.Range(0, 0xFF)
2 . .Select(i => myNamespace + (char)i + "." + "MyTypeName")
3 . .Where(n => System.Reflection.Assembly.Load(myAssembly).GetType(n) != null)
1 Enumerable.WhereEnumerableIterator<string> { "System.Data.Services.Internal.ExpandedWrapper`1[[System.DateTime, mscorlib,
Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089]]\0.MyTypeName" }
1 >
```

Surprisingly, a null character `\0` appears to terminate the parsing of the type name and the appendix `.MyTypeName` gets ignored! What year is it, again!?

Server control tag names can only be `[\w:~.]+`

Check. As the appended type name gets ignored it can be chosen arbitrarily.

And because a null character can be appended to the `Namespace` attribute of the `@Register` directive, we can specify a custom server control of type `ExpandedWrapper<DateTime>` and set its `ExpandedElement` property:

```
<%@ Register
    TagPrefix="x"
    Assembly="System.Data.Services, Version=4.0.0.0, Culture=neutral,
    PublicKeyToken=b77a5c561934e089"

    Namespace="System.Data.Services.Internal.ExpandedWrapper`1[[System.DateTime,mscorlib]]\0"
%>
<x:y
    runat="server"
    ExpandedElement="foobar"
/>
```

Great! So this works — at least out of the context of SharePoint's `SPPageParserFilter`:

**Request**

RawParamsHeadersHex

POST /Controls/TemplateControl.aspx HTTP/1.0  
Host: localhost  
Content-Type: multipart/form-data; boundary=-----2030902359  
Content-Length: 394  
  
-----2030902359  
Content-Disposition: form-data; name="control"  
  
<%@ Register  
TagPrefix="x"  
Assembly="System.Data.Services, Version=4.0.0.0, Culture=neutral, PublicKeyToken=b77a5c561934e089"  
Namespace="System.Data.Services.Internal.ExpandedWrapper`1[[System.DateTime,mscorlib]]&#0;"  
%>  
<x:y  
runat="server"  
ExpandedElement="foobar"  
/>  
  
-----2030902359--

?<+>Type a search term

0 matches

**Response**

RawHeadersHexHTMLRender

## Server Error in '/' Application.

---

### Parser Error

**Description:** An error occurred during the parsing of a resource required to service this request. Please review the following specific parse error details and modify your source file appropriately.

**Parser Error Message:** Cannot create an object of type 'System.DateTime' from its string representation 'foobar' for the 'ExpandedElement' property.

**Source Error:**

|         |                                                                                             |
|---------|---------------------------------------------------------------------------------------------|
| Line 4: | Namespace="System.Data.Services.Internal.ExpandedWrapper`1[[System.DateTime,mscorlib]]&#0;" |
| Line 5: | %>                                                                                          |
| Line 6: | <x:y                                                                                        |
| Line 7: | runat="server"                                                                              |
| Line 8: | ExpandedElement="foobar"                                                                    |

The remaining challenge was to find a generic type that passed the `SPPageParserFilter`. As well as a targeted type that provides an interesting static `Parse(string)` method or has associated an interesting `TypeConverter` that gets called on property setting. And there sure was at least one in SharePoint:

```
https://sharepoint.com/

Command Line: c:\windows\system32\cmd.exe -ap "SharePoint Content Services" -v "v4.0" -l "webengine.dll" -a "\\pipe\\llslpmb7025d08-d572-4b30-94ef-23136eaffa0b" -h "C:\inetpub\temp\lappools\SharePoint Content Services\SharePoint Content Services.config" -w ""
User: NT AUTHORITY\SYSTEM
Environment Variables:
MONITORING_CIL_HRM_ACCOUNT=005P_CilInternal
RoleName=USR
MONITORING_FDS_NAMESPACES=nsSPOProd
OS=Windows_NT
MONITORING_OBJECTSTORE_ENDPOINTS=NotSet
ProgramData=C:\ProgramData
PSPoolPath=C:\Program Files\WindowsPowerShell\Modules\C:\Windows\system32\WindowsPowerShell\v1.0\Modules;c:\Program Files\AppFabric 1.1 for Windows Server\PowershellModules
CpInstallPath=C:\CPA\1.22.505.3\drop\CPA\1.22.504.3\Agent
MONITORING_ROLE=USR
MONITORING_TRACELOG_DIRECTORY=C:\SPOCHA_V2\Bin\TraceLogs
MONITORING_DATA_DIRECTORY=D:\SPOCHA_V2\Data
MONITORING_FARM_ID=
COR_ENABLE_PROFILING=1
MONITORING_OCS_THUMBPRINT=
InterceptExtensionDirectory=C:\Intercept\
MONITORING_CONFIG_VERSION=16.37
ComSpec=C:\Windows\system32\cmd.exe
USERHOSTNAME=SP-uahm0004.HSFT.net
MONITORING_ULS_LOGDIR=D:\Logs\ULS
MONITORING_USE_GENEVA_CONFIG_SERVICE=True
ASSEMBLY_ENABLED_SCENARIOS=CertInUseError,CertInUseEvent
PROCESSOR_REVISION=5504
MONITOR_DIMENSION_Machine=
NUMBER_OF_PROCESSORS=16
MONITORING_ALTERNATE_HRM_ACCOUNTS=ServiceInternalQosEvent,SPOProdInternal\ServiceIncomingQosEvent,SPOProdIncoming\ServiceOutgoingQosEvent,SPOProdOutgoing\SPMonitoredScopeExitEvent,SPOProdULS\QServiceInternalQosEvent,QProd\QrIdClientQosEvent,QProd\QrIdInternalC
ProgramData=C:\Program Files
PROCESSOR_LEVEL=0
MONITORING_OCS_NAMESPACES=nsSPOProd
CurrentSPOVersionId=4.23529.12003
MONITORING_NETWORK=
MONITORING_AGENT_CONFIG_NAME=SPOContent
DriverData=C:\Windows\system32\Drivers\DriverData
MONITOR_IGNORE_BUILT_IN_DIMENSION=1
COR_PROFILER_PATH_64=C:\Intercept\vd4\MicrosoftInstrumentationEngine_x64.dll
MONITORING_GROVESTORY=0_h0b_0
COMPUTERNAME=US8188580-594
MONITORING_ROLE_INSTANCE=
RoleInstance=
MONITORING_IDENTITY=use_ip_address
EnvironmentName=QProdQ004
CommonProgramFiles(x86)=C:\Program Files (x86)\Common Files
TMP=C:\Users\SHAREP-1\AppData\Local\Temp
MONITOR_DIMENSION_FarmId=
MONITORING_DATACENTER=JMS22
CORPlus.ThreadPool.UnfairSemaphoreSpinLimit=0
MONITOR_SERVICES=SPTracerV4,CircuitBreaker,ULSControllerService,SPTraceV4,HSSQLSERVER,NetPipeActivator,MSOfficeDataLoader,MS
MONITORING_FARM_LABEL=EHEA_275_CONTENT
ULS_EXPAND_FORCED_LOGGING_MESSAGES=True
MONITORING_OCS_CERTSTORE=LOCAL_MACHINE_WY
MONITORING_OCS_REGION=WestEurope
MONITOR_DIMENSION_FarmLabel=EHEA_275_CONTENT
MONITORING_RESOURCEFQDN=QProdQ004.HSFT.NET
MONITOR_DIMENSION_Role=USR
PUBLIC=C:\Users\Public
MONITORING_REGION=WestEurope
windir=C:\Windows
MONITORING_OCS_ACCOUNT=SPOProd
DiagnosticStore=D:\SPOCHA_V2\data
SPOVersion=16.4.23529.12003/4/10/2023 5:40:02 PM
PROCESSOR_IDENTIFIER=Intel64 Family 6 Model 85 Stepping 4, GenuineIntel
SERVICE_TREE_ID=
APPDATA=C:\Users\SharePoint Content Services\AppData\Roaming
MONITORING_ISACOREBY=Primary
Path=C:\Program Files\Microsoft Office Servers\16.0\Search\Native\jic\Program Files\Common Files\Microsoft Shared\ULS\16\jic\Program Files\Common Files\Microsoft Shared\Web Server Extensions\16\Bin\jic\Program Files\Microsoft Office Servers\16.0\Bin\jic\Program
COR_PROFILER=
COR_PROFILER_PATH_64=C:\Intercept\vd4\MicrosoftInstrumentationEngine_x64.dll
MONITORING_AGENT_CERTTYPE=SPOCHA
PATHEXT=.COM;.EXE;.BAT;.CMD;.VBS;.VBE;.JS;.JSE;.WSF;.WSH;.MSC
INFI\APPDATA\Local\SharePoint Content Services\LocalData\Local
```

We won't publish the exploit here and leave it as an exercise to the reader.

# The Patch

Microsoft fixed this vulnerability ([CVE-2023-33160](#)) in the July 2023 security updates for SharePoint by also checking the type arguments of a generic type. Here is the diff of the decompiled `SPageParserFilter` class files of the assembly file versions 16.0.16130.20548 (June 2023) and 16.0.16130.20640 (July 2023):

```

diff --git
a/Microsoft.SharePoint/Microsoft/SharePoint/ApplicationRuntime/SPPageParserFilter.cs
b/Microsoft.SharePoint/Microsoft/SharePoint/ApplicationRuntime/SPPageParserFilter.cs
index d1ee368..7f893c5 100644
--- a/Microsoft.SharePoint/Microsoft/SharePoint/ApplicationRuntime/SPPageParserFilter.cs
+++ b/Microsoft.SharePoint/Microsoft/SharePoint/ApplicationRuntime/SPPageParserFilter.cs
@@ -146,9 +146,32 @@ namespace Microsoft.SharePoint.ApplicationRuntime
     {
         flag = false;
     }
-    else if
+    ((this.IsTypeOrSubclass(typeof(Control), controlType) ||
+!this._safeModeDefaults.ControlCompatMode) &&
+!this._pageParserSettings.AllowUnsafeControls)
+    else if
+    ((this.IsTypeOrSubclass(typeof(Control), controlType) ||
+!this._safeModeDefaults.ControlCompatMode) &&
+(!this._pageParserSettings.AllowUnsafeControls ||
+AllowUnsafeControlsOverride.AlwaysBlockUnsafeControls))
     {
         flag =
this._safeControls.IsSafeControl(this._isAppWeb, controlType, out text);
+        if (flag)
+        {
+            foreach (Type type
+in controlType.GetGenericArguments())
+            {
+                try
+                {
+                    flag = this.AllowControl(type, childBuilder);
+                }
+                catch
+                {
+                    ULS.SendTraceTag(539066894U, ULSCat.msoulscat_WSS_Runtime, ULSTraceLevel.Medium,
+                    "AllowControl failed on templateType: {0} in ControlType VirtualPath: {1}", new object[]
+                    {

```

```
+ type,
+
controlType
+
});
+
throw;
+
}
+
}
+
}
+
if (!flag &&
this.IsTypeOrSubclass(typeof(UserControl), controlType) &&
this._allowedUserControlVirtualPaths != null)
{
    if
(
(this._allowUserControlTypes == null)
@@ -182,7 +205,7 @@ namespace Microsoft.SharePoint.ApplicationRuntime
return false;
}
string text2;
if
(!this._pageParserSettings.AllowUnsafeControls &&
!this._safeControls.IsSafeControl(this._isAppWeb, controlType, out text2))
+
if
((!this._pageParserSettings.AllowUnsafeControls ||
AllowUnsafeControlsOverride.AlwaysBlockUnsafeControls) &&
!this._safeControls.IsSafeControl(this._isAppWeb, controlType, out text2))
{
ULS.SendTraceTag(2744449U, ULSCat.msoulscat_WSS_Runtime, ULSTraceLevel.High, "Allowing
ControlCompatMode=true class on page. Class: {0}, VirtualPath: {1}, TagName: {2}", new
object[]
{
@@ -226,7 +249,7 @@ namespace Microsoft.SharePoint.ApplicationRuntime
{
    this._IsMasterPage =
this.IsTypeOrSubclass(typeof(MasterPage), baseType);
    string text = null;
-
bool flag = this.Exclusion ||
this._pageParserSettings.AllowUnsafeControls ||
this._safeControls.IsSafeControl(this._isAppWeb, baseType, out text);
+
bool flag = this.Exclusion ||
(this._pageParserSettings.AllowUnsafeControls &&
```

```

!AllowUnsafeControlsOverride.AlwaysBlockUnsafeControls) ||
this._safeControls.IsSafeControl(this._isAppWeb, baseType, out text);
        if (!flag && text != null)
        {
            throw new
SafeControls.UnsafeControlException(SPResource.GetString("UnsafeBaseTypePageParserFilterE
rror", new object[]
@@ -253,7 +276,7 @@ namespace Microsoft.SharePoint.ApplicationRuntime
        case VirtualReferenceType.Page:
            return
!SPRequestModule.IsExcludedPath(referenceVirtualPath, false);
        case VirtualReferenceType.UserControl:
            flag =
(this._pageParserSettings.AllowUnsafeControls ||
this._safeControls.IsSafeControl(this._isAppWeb, referenceVirtualPath));
            +
            flag =
((this._pageParserSettings.AllowUnsafeControls &&
!AllowUnsafeControlsOverride.AlwaysBlockUnsafeControls) ||
this._safeControls.IsSafeControl(this._isAppWeb, referenceVirtualPath));
            if (flag)
            {
                if
(this._allowedUserControlVirtualPaths == null)

```

Note that this patch only applies to SharePoint and not to ASP.NET in general. So if you have control over the code to be parsed by `TemplateParser`, you still can achieve Remote Code Execution.

## Conclusion

The `TemplateParser` of ASP.NET holds unexpected capabilities, which can lead to Remote Code Execution during the parsing process.

Special thanks are due to the null character truncation that enables the use of generic types. This truncation also applies to assembly and type loading in general, which may allow bypassing security protections such as serialization binders (see also [Bypassing .NET Serialization Binders](#)).

In fact, it appears that even currently supported .NET runtimes are still affected:

| Runtime | Assembly Name Truncation | Type Name Truncation |  | .NET Framework 4.8.1 |
|---------|--------------------------|----------------------|--|----------------------|
| ✓       | ✓                        |                      |  | .NET 6               |
| ✓       | ✓                        |                      |  | .NET 7               |
| □       | ✓                        |                      |  | .NET 8               |
| □       | ✓                        |                      |  |                      |

The vulnerabilities in Sitecore ([CVE-2023-35813](#)) and SharePoint ([CVE-2023-33160](#)) demonstrate that vulnerabilities utilizing the ASP.NET `TemplateParser` are not just theoretical but have an actual impact on popular real world applications. And due to the fact that assembly and type name truncation still works in current .NET releases, it is likely that other vulnerable software is out there waiting to be found.

